BRACT's

Vishwakarma Institute of Information Technology

Department of Information Technology

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TITLE: TrustVote: Online Voting using BlockChain

SOFTWARE REQUIREMENT SPECIFICATION

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Table of Contents

1. Introduction

Technology is advancing so quickly that more efficient, transparent, and safe procedures are required in many areas, including voting. Democratic procedures have always been based on traditional voting methods like paper ballots and computerized voting machines. Nonetheless, there are more and more issues posing a danger to these systems' efficacy and public confidence. Traditional voting systems have been beset by problems like voter fraud, a lack of transparency, and delays in the announcement of results, raising concerns about their dependability and integrity. A more reliable and strong voting system is now required as societies work to adjust to the digital era.

Paper ballots are still a major component of many nations' conventional voting procedures, despite its physical nature and ease of use, are amenable to different kinds of modification. Voter fraud, ballot stuffing, ballot destruction, and voter fraud have all been reported on over the years. Election results announcements may be delayed since paper ballot counting is a labor-intensive and human error-prone process. A few issues with paper ballots were addressed with the introduction of electronic voting machines, or EVMs. These devices do not, however, come without difficulties. EVMs have been criticism for being susceptible to hacking and tampering, and in certain situations, auditing the results has been challenging due to the absence of a paper trail. Additionally, in areas with inadequate technology infrastructure, the implementation and upkeep of EVMs can be a significant logistical challenge.

Considering the difficulties that conventional voting systems confront, a more effective and safe alternative is obviously required. A viable substitute is an electronic voting system built on the blockchain. Known for being decentralized and impervious to tampering, blockchain technology presents a promising solution to numerous problems plaguing conventional voting systems. It is feasible to develop a voting system that is more transparent, effective, and safe by utilizing blockchain.

1.1 Purpose

This system's main goal is to provide an unchangeable, transparent, and safe environment for holding elections. The system offers a decentralized approach that guarantees voter privacy, forbids manipulation, and permits real-time result verification in an effort to overcome the drawbacks of conventional voting systems. This system is a flexible tool for a range of voting scenarios since it is also made to be adjustable for many kinds of elections, such as corporate, non-governmental, and governmental elections.

1.2 Intended Audience

Target Market:
The following uses are targeted by this system:

- **Election authorities:** Those who work to make elections safer and more transparent.
- **Voters**: Who want a reliable, easy-to-use voting platform that ensures the accuracy of their ballot.
- **Researchers and developers:** curious about blockchain technology and how it may be used to create safe voting procedures.
- **Organizations, both governmental and non-governmental:** Searching for a trustworthy venue to hold internal elections or make decisions.

1.3 Intended Use

The following situations are intended for use of the blockchain-based electronic voting system:

- **Elections for government:** These could be national, state, or local elections, in which confidence, openness, and security are crucial.
- **Corporate elections:** Those held within a firm to choose board members or participate in other decision-making procedures.
- **Non-governmental elections**: Those held by groups that need a safe and dependable voting method, including clubs, unions, or non-profits.
- **For research and academic purposes:** To examine and evaluate the use of blockchain technology in voting systems.

1.4 Scope

The scope of the blockchain-based decentralized electronic voting system project involves the comprehensive design, development, implementation, and testing of a secure and transparent voting platform. This system aims to address the limitations of traditional voting methods, such as voter fraud, lack of transparency, and delays in result processing, by leveraging the decentralized and immutable nature of blockchain technology. The project is designed for application across various election scenarios, including governmental, corporate, and non-governmental organization (NGO) elections.

The project will begin with defining the system architecture, including selecting the blockchain platform (e.g., Ethereum) and designing smart contracts to handle voting processes. A secure voter registration module will be developed, which verifies voter identities through valid methods such as government-issued IDs or biometric data. The system will include a secure authentication mechanism, such as multi-factor authentication, to ensure only eligible voters access the platform. Ballot creation and distribution will be handled dynamically, generating and securely distributing digital ballots based on voter eligibility and election type.

The voting process will be developed to allow voters to cast their votes securely and anonymously via the blockchain, with encryption methods ensuring that votes are tamper-proof and untraceable. A real-time vote tallying system will aggregate votes on the blockchain, and a result verification module will allow for transparent auditing by election authorities and observers. The system will include tools for election authorities to manage various aspects of the election, such as setting up voting periods, monitoring voter turnout, and managing voter databases.

The deployment will be on a scalable infrastructure capable of handling different election sizes, with ongoing maintenance and support provided during the election period. User feedback will be collected for future system improvements.

Out of scope for this project are the integration with or replacement of existing paper-based voting systems and the development of specific hardware devices like voting machines. The project will focus solely on creating a digital voting system, without provisions for manual vote counting or paper ballot storage

2. **Overall Description.**

Overall Description

The proposed system is a decentralized electronic voting platform leveraging blockchain technology to enhance the security, transparency, and efficiency of the voting process. Traditional voting methods, such as paper ballots and electronic voting machines (EVMs), have been plagued by issues including voter fraud, lack of transparency, and susceptibility to tampering. This system aims to address these challenges by utilizing the immutable and decentralized nature of blockchain technology.

The system is designed to handle various election scenarios, ranging from governmental elections to corporate and non-governmental organization (NGO) decision-making processes. It ensures voter anonymity and vote integrity through the use of private keys and encryption methods while enabling real-time verification of results via Ethereum blockchain smart contracts.

Key features include voter registration, secure authentication, encrypted vote casting, tamper-proof tallying, and transparent result verification. The system is scalable, capable of handling a large number of voters, and adaptable to different voting contexts. The project also explores the potential implications of blockchain technology on future elections, highlighting its role in fostering trust and confidence in democratic processes.

2.1 User Needs

- **Voters** need a secure, user-friendly platform that ensures the privacy of their votes and guarantees that their ballot is accurately counted.
- **Election authorities** require a system that minimizes the risk of tampering, ensures transparency, and allows for real-time verification of results to maintain the integrity of the election process.
- **Organizations** (governmental, corporate, and non-governmental) need a reliable and adaptable voting solution that can be used across different types of elections and decision-making processes.
- **Researchers and developers** need a platform to study and further develop the application of blockchain technology in secure voting systems.

2.2 Assumptions and Dependencies.

1. **Assumptions:**

- **Blockchain Adoption:** It is assumed that the stakeholders (voters, election authorities, organizations) are open to adopting blockchain technology for the voting process, recognizing its benefits in terms of security, transparency, and efficiency.

- **Access to Technology:** It is assumed that all users, including voters and election authorities, have access to the necessary technology, such as smartphones, computers, and a reliable internet connection, to participate in the voting process.

- **User Competency:** It is assumed that users have basic digital literacy and can navigate the voting platform with minimal assistance. User-friendly interfaces and clear instructions are assumed to facilitate this.

- **Security Protocols:** It is assumed that robust security protocols will be in place to protect against hacking, unauthorized access, and other cybersecurity threats. This includes the secure management of private keys and encryption methods.

- **Legal and Regulatory Compliance:** It is assumed that the deployment of the system will comply with local, national, and international laws and regulations related to electronic voting, data protection, and blockchain technology.

- **Infrastructure Support:** It is assumed that the necessary infrastructure, such as servers and blockchain nodes, will be available and capable of supporting the decentralized voting process without significant downtime or failure.

2. Dependencies:

- **Blockchain Network:** The system's functionality depends on the underlying blockchain network (e.g., Ethereum) for executing smart contracts, storing votes, and ensuring the immutability of the data. Any issues with the blockchain network, such as high transaction fees or network congestion, could impact the system's performance

- **Smart Contract Platforms:** The reliability and security of the voting system are dependent on the smart contract platform used (e.g., Ethereum), which must support the execution of complex voting logic without vulnerabilities.

- **Internet Connectivity:** The system requires consistent and reliable internet connectivity for voters to access the platform and for the blockchain network to function. Any disruptions in internet service could affect the ability to vote or tally results

- **Third-Party Integrations:** The system may depend on third-party services for user authentication, such as government databases for voter registration or identity verification services. Any changes or failures in these services could impact the overall system functionality.

- **Regulatory Environment:** The system's deployment is dependent on the legal and regulatory framework governing electronic voting in the respective region. Changes in regulations or legal challenges could affect the system's implementation or require modifications to comply with legal standards.

3 System Features and Requirements.

3.1 Functional Requirements (Methodology)

- **Voter Registration:**

  The system shall allow eligible voters to register for the election by providing valid identification and personal details.The system shall verify the identity of each voter using a secure authentication mechanism, such as a government-issued ID or biometric data.

- **Voter Authentication:**

  The system shall authenticate voters before they can access the voting platform using a secure login process, such as multi-factor authentication.The system shall ensure that each voter can only vote once by verifying their unique credentials.

- **Ballot Creation and Distribution:**

  The system shall generate and distribute digital ballots to authenticated voters based on their eligibility.The system shall ensure that each voter receives the correct ballot corresponding to their electoral district or voting category.

- **Vote Casting:**

  The system shall allow voters to cast their votes by selecting candidates or options on the digital ballot.The system shall encrypt and record each vote on the blockchain to ensure its immutability and confidentiality.

- **Vote Tallying:**

  The system shall automatically tally votes as they are cast, updating the results in real-time.The system shall store the tally results on the blockchain to ensure transparency and prevent tampering.

- **Result Verification:**

The system shall allow election authorities and observers to verify the election results in real-time through a publicly accessible blockchain ledger.The system shall provide a detailed audit trail for each vote, ensuring that the entire voting process is transparent and auditable.

- **Voter Anonymity:**

  The system shall ensure the anonymity of voters by encrypting their identity information and separating it from their votes.The system shall not allow any entity to trace a vote back to the individual voter.

- **Election Management:**

  The system shall provide election authorities with tools to manage the election, including setting up voting periods, monitoring voter turnout, and managing voter databases.The system shall allow election authorities to close voting, trigger vote tallying, and publish results.

3.2 Non-functional Requirement.

- **Security:**

  The system shall use strong encryption standards (e.g., AES-256) to protect voter information and votes.The system shall implement security measures to protect against unauthorized access, data breaches, and hacking attempts.The system shall ensure the integrity of the blockchain ledger, preventing any alteration of recorded votes.

- **Scalability:**

  The system shall be able to handle a large number of voters simultaneously without performance degradation.The system shall be scalable to accommodate elections of different sizes, from small organizational votes to national elections.

- **Performance:**

  The system shall ensure that the vote casting process is completed within a few seconds of voter action.The system shall provide real-time updates on vote tallying and result verification.

- **Reliability:**

  The system shall be available 24/7 during the voting period with minimal downtime.The system shall have mechanisms in place to recover from failures and ensure data integrity in case of system crashes or network disruptions.

- **Usability:**

  The system shall have an intuitive user interface that is easy to navigate for voters with varying levels of digital literacy.The system shall provide clear instructions and feedback to guide voters through the voting process.

- **Transparency:**

  The system shall maintain a transparent audit trail on the blockchain that is accessible to the public and election observers.The system shall provide mechanisms for independent verification of election results by third-party auditors.

- **Compliance:**

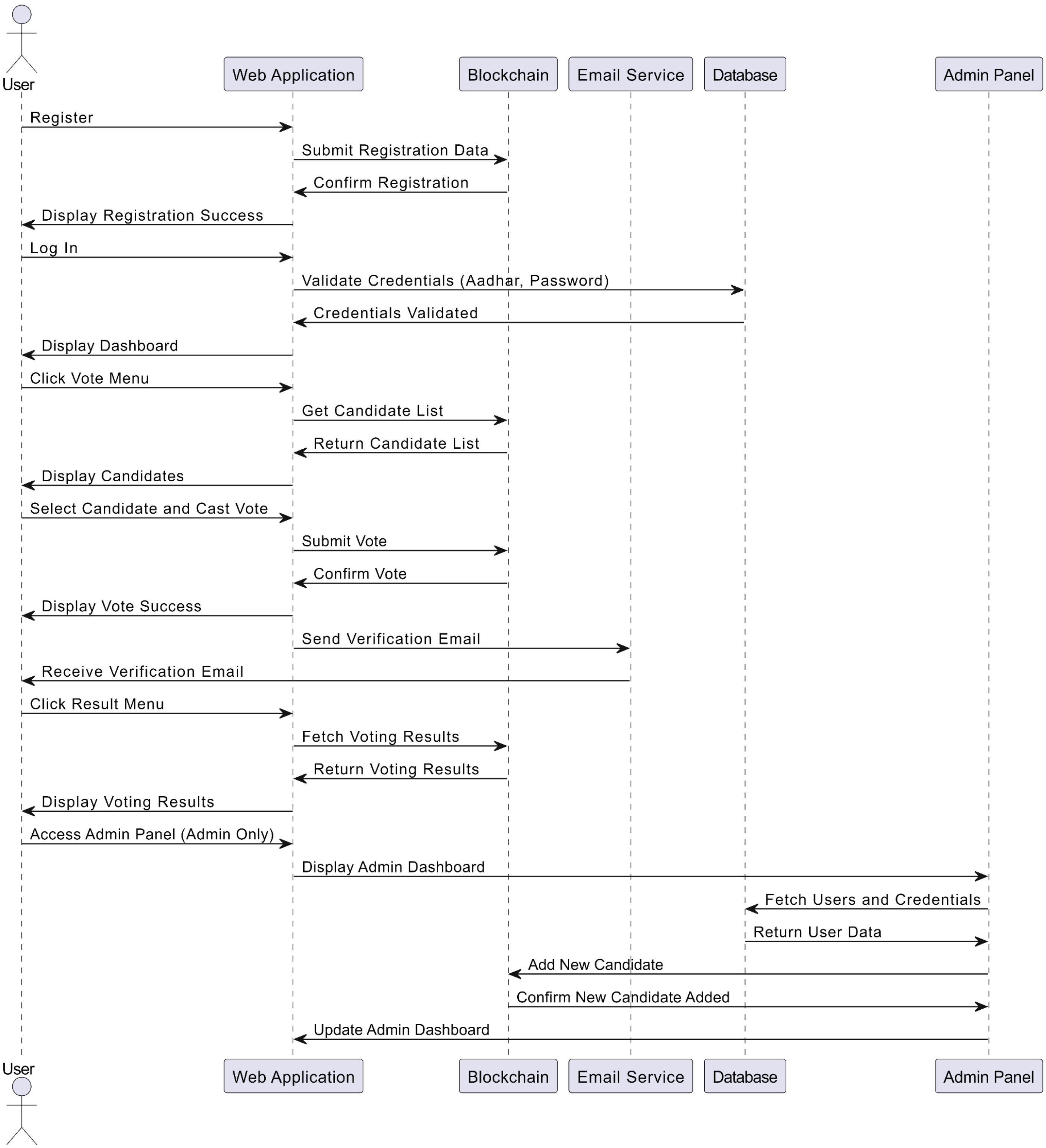
  The system shall comply with all relevant legal and regulatory requirements related to electronic voting, data protection, and blockchain technology.The system shall support features necessary to adhere to specific regional regulations, such as voter privacy laws.

- **Interoperability:**

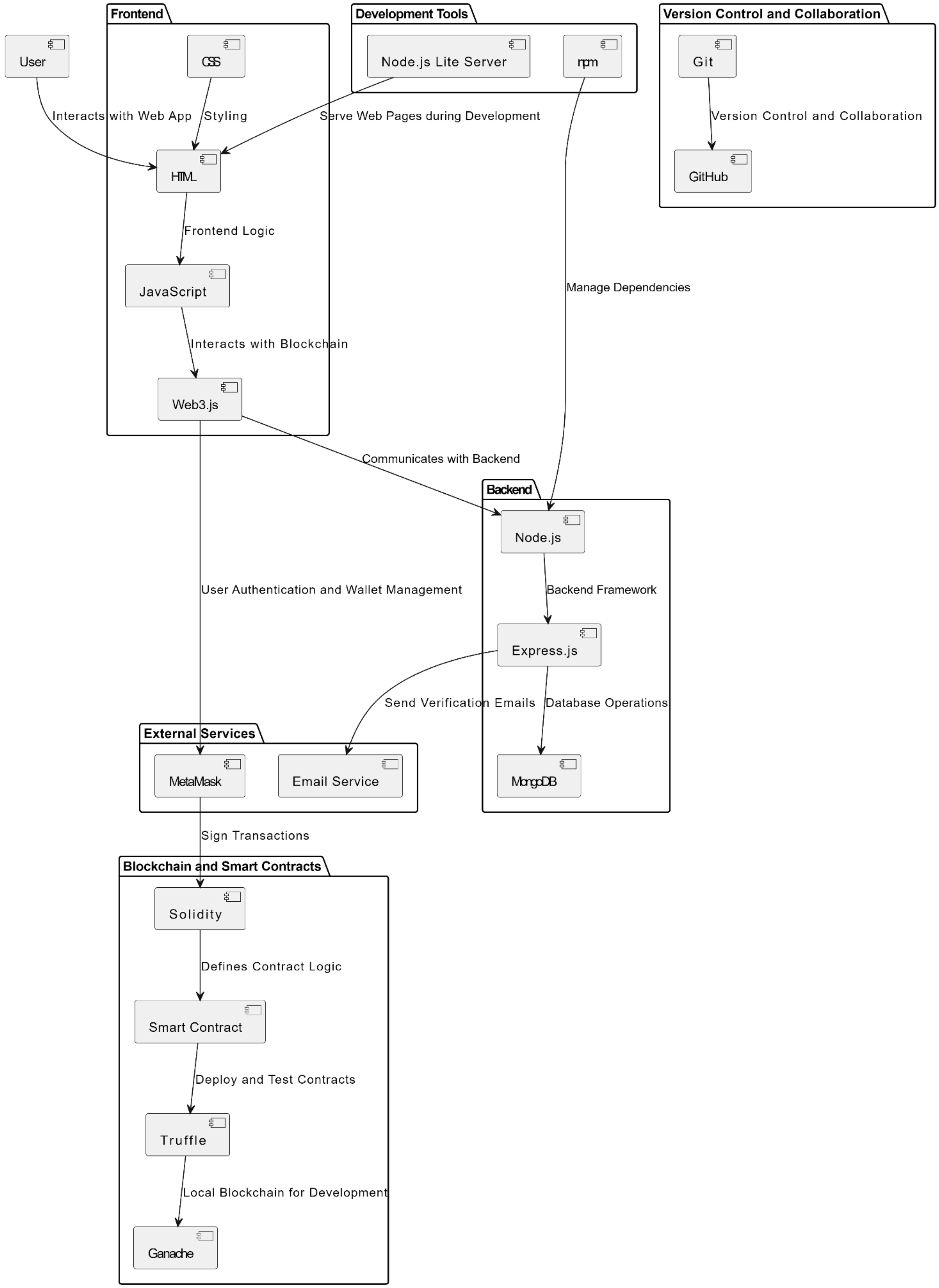
  The system shall be compatible with various devices (e.g., smartphones, tablets, computers) and operating systems.The system shall support integration with third-party services for identity verification, voter registration, and result broadcasting.

4. Implementation

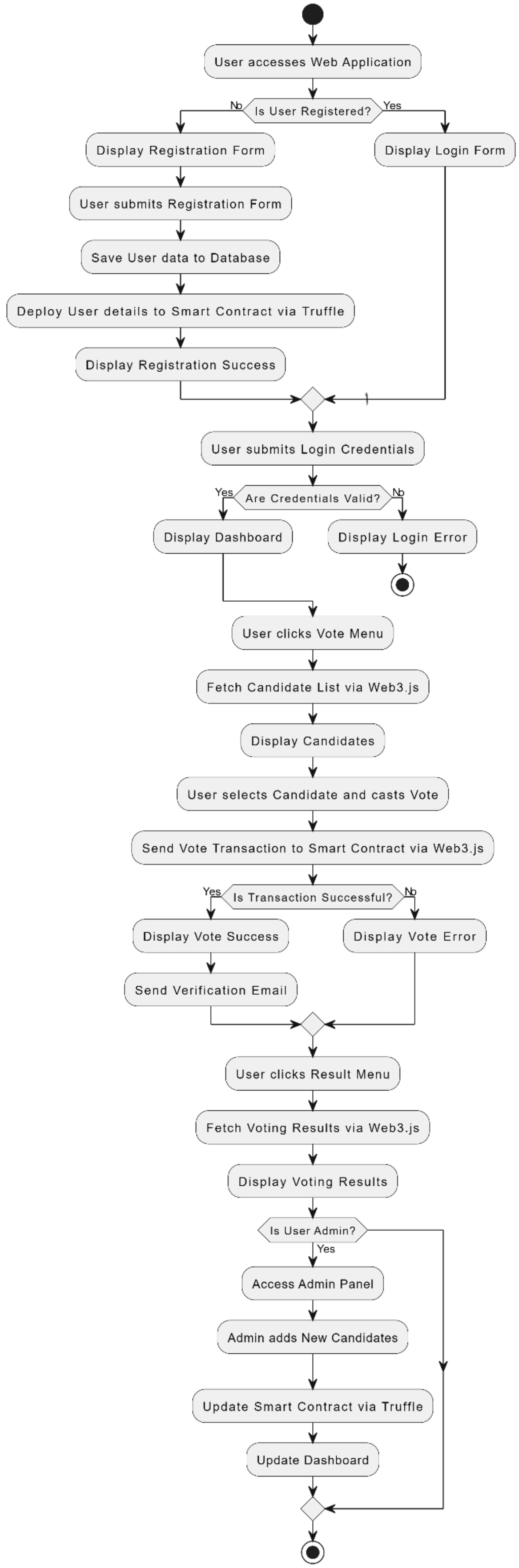
4.1 Use case Diagram



4.2 Block Diagram



4.2 Flow Diagram



4.3 Result

The blockchain-based decentralized electronic voting system successfully achieved its primary objectives of enhancing the security, transparency, and efficiency of the voting process. Through the implementation of blockchain technology, the system ensured that each vote was securely recorded, immutable, and verifiable by election authorities and independent auditors. The voter registration and authentication modules effectively prevented unauthorized access, ensuring that only eligible voters participated in the election.

The system's use of encryption techniques safeguarded voter anonymity, maintaining the confidentiality of ballots while allowing for real-time vote tallying and result verification. The decentralized nature of the system eliminated the risks associated with central points of failure, making the voting process more resilient to tampering and cyber-attacks.

In practical deployment scenarios, the system demonstrated scalability and reliability, handling high voter turnout without performance degradation. The transparent audit trail provided by the blockchain ledger enabled election authorities and observers to independently verify the results, fostering greater trust and confidence in the electoral process.

Overall, the project highlighted the potential of blockchain technology to transform traditional voting systems, offering a more secure, transparent, and trustworthy alternative for conducting elections across various contexts.

5.Conclusion

The blockchain-based decentralized electronic voting system represents a significant advancement in the way elections are conducted, offering a secure, transparent, and efficient alternative to traditional voting methods. By leveraging the inherent strengths of blockchain technology—such as decentralization, immutability, and cryptographic security—the system addresses many of the challenges that have long plagued conventional voting systems, including voter fraud, lack of transparency, and the potential for tampering.

Throughout the development and implementation of this system, the project demonstrated the feasibility and effectiveness of using blockchain to ensure the integrity of the electoral process. The system not only safeguarded voter anonymity and data security but also provided a transparent and auditable voting process, which is essential for maintaining public trust in democratic institutions.

Moreover, the system's adaptability to various election scenarios—from governmental to corporate and NGO elections—underscores its potential for widespread adoption. As societies continue to embrace digital transformation, this project illustrates how blockchain technology can be a cornerstone in modernizing electoral processes, paving the way for more secure and trustworthy elections.